

COVER PAGE

IGRVT-50 Payload EE

Schematic

Version: 1.0.0



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

Sheet title: CoverPage.SchDoc

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Checked by: BAO BUI

Document No: 1

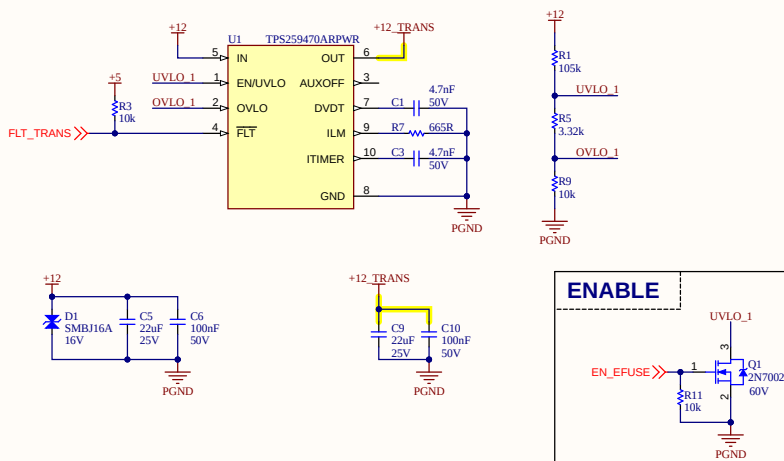
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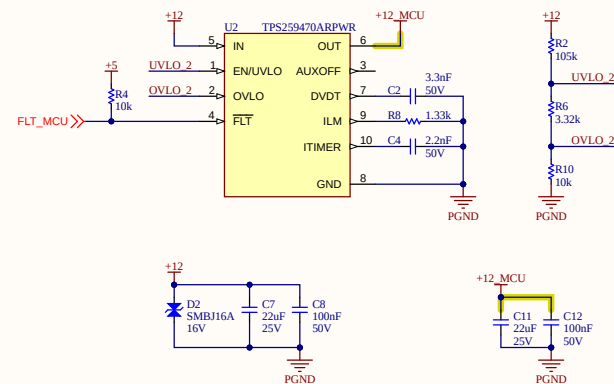
Revision: 1.0.0

Date: 8/10/2026

eFuse_Transducer



eFuse_MCU



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

Sheet title: eFuse.SchDoc

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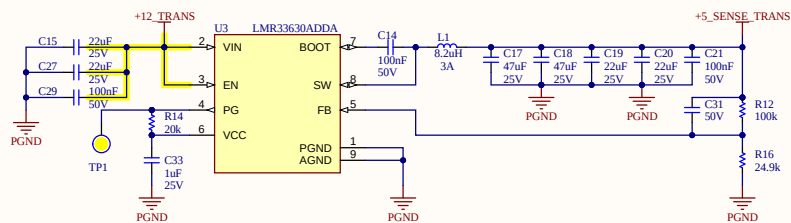
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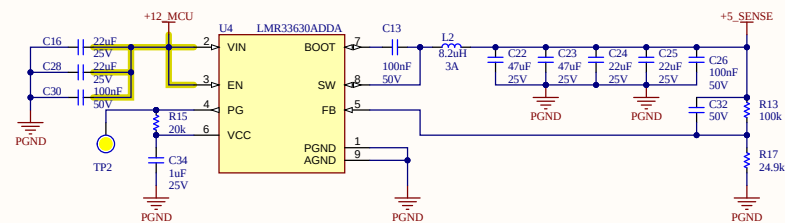
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5V TRANSDUCER



5V MCU



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

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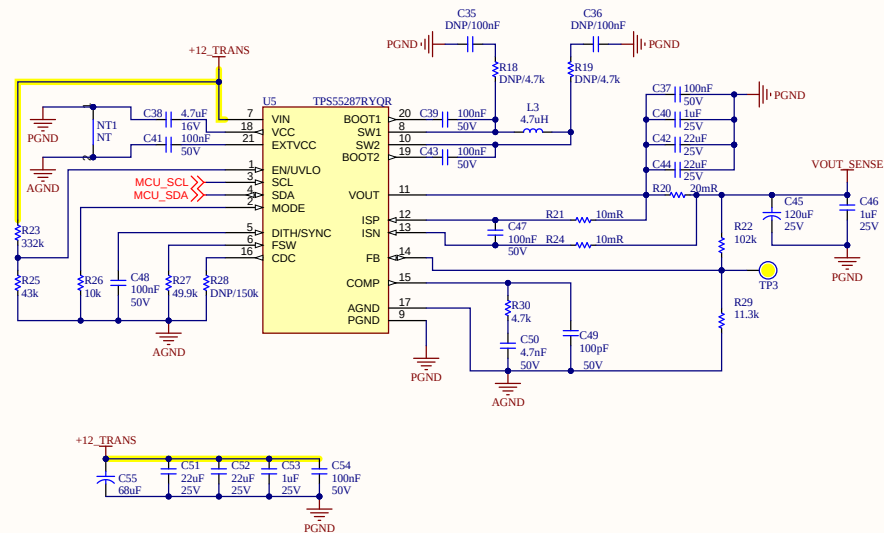
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Revision: 1.0.0

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5-12V BUCK



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

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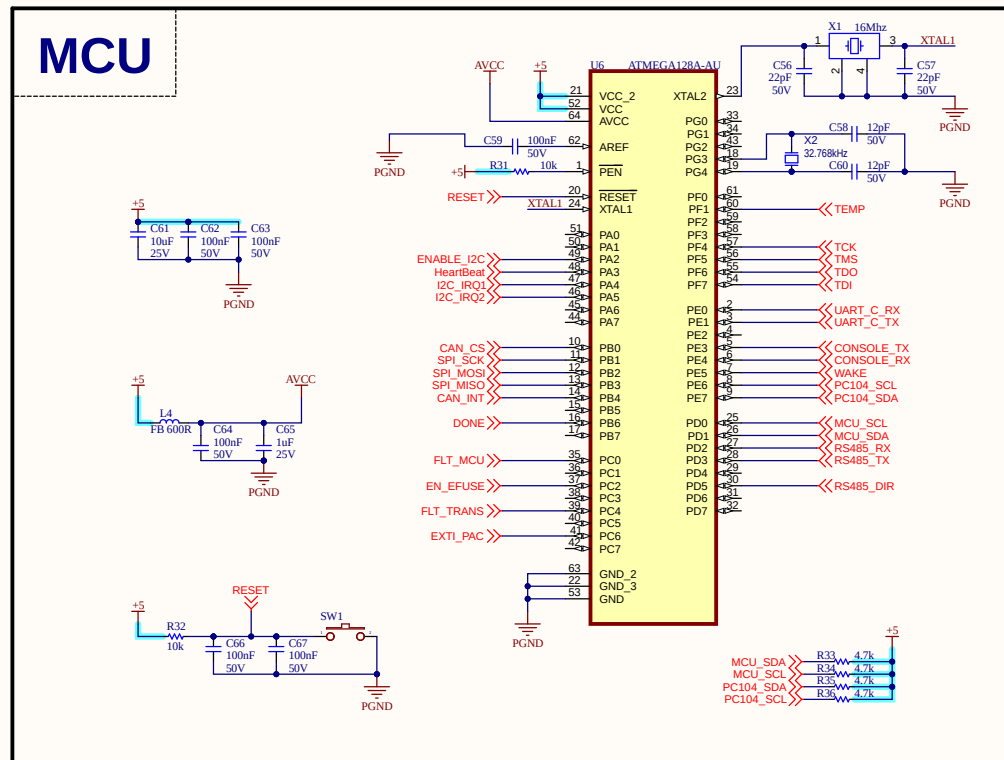
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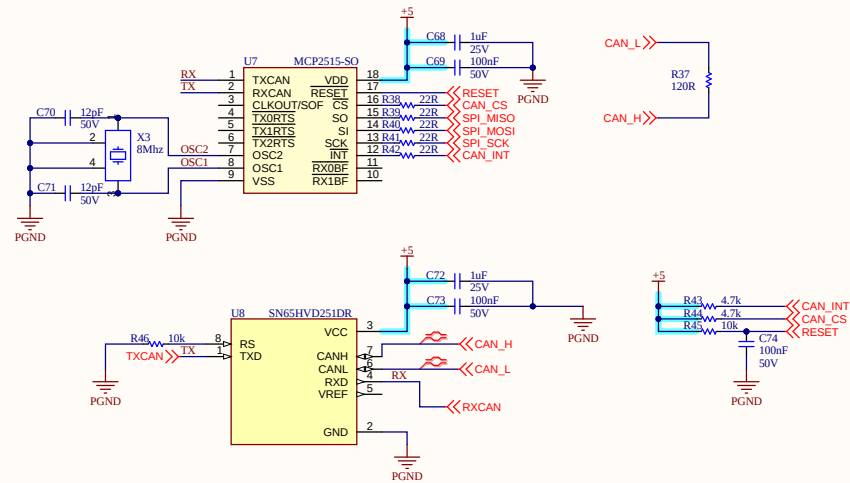
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CAN



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RS485

The diagram illustrates the electrical connection for an RS485 interface using a MAX485ESA+ transceiver. The transceiver is represented by a yellow rectangular block labeled U9. Its pins are connected as follows:

- Pin 1 (VCC):** Connected to a +5V supply through a 10kΩ resistor (R47).
- Pin 2 (RE):** Connected to the RS485_DIR signal.
- Pin 3 (DE):** Connected to the RS485_TX signal.
- Pin 4 (DI):** Connected to the RS485_A signal.
- Pin 5 (GND):** Connected to a common ground (PGND).
- Pin 6 (A):** Connected to the RS485_A signal.
- Pin 7 (B):** Connected to the RS485_B signal.
- Pin 8 (RO):** Connected to the RS485_RX signal through a 10kΩ resistor (R48).

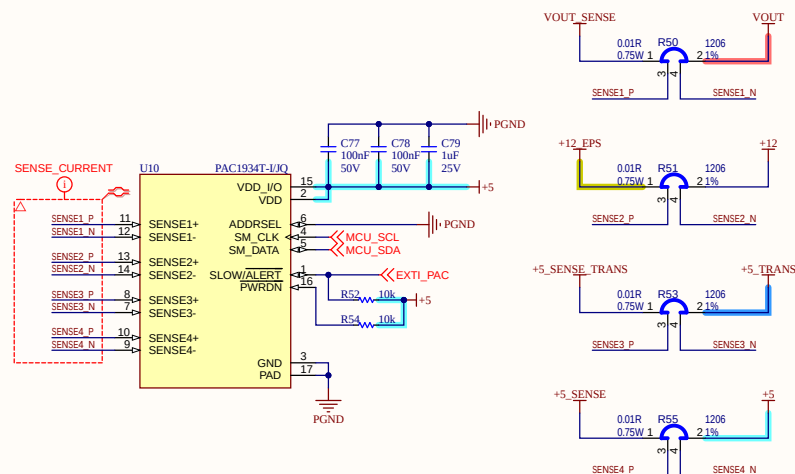
On the left, the RS485_A and RS485_B signals are shown as differential pairs. RS485_A is connected to a 120Ω resistor (R48) in series with a 10kΩ resistor (R47) to the +5V supply. RS485_B is connected to a 10kΩ resistor (R48) to ground (PGND).

On the right, the RS485_RX signal is shown as a differential pair. It is connected to the RO pin (pin 8) of the transceiver through a 10kΩ resistor (R48). The RS485_RX signal is also connected to a 10kΩ resistor (R47) to the +5V supply and a 100nF capacitor (C75) to ground (PGND). The RS485_TX signal is connected to the DE pin (pin 3) of the transceiver through a 10kΩ resistor (R48).



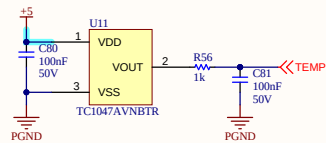
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Design by: KIEN HOANG	Sheet title: RS485.SchDoc	Sheet 7 of 15
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POWER MONITORING



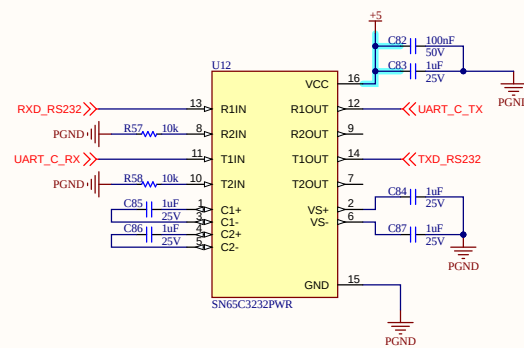
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Design by: KIEN HOANG	Sheet title: PowerMonitor.SchDoc	Sheet 8 of 15
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TEMPERATURE SENSOR



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RS232



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

Sheet title: RS232.SchDoc

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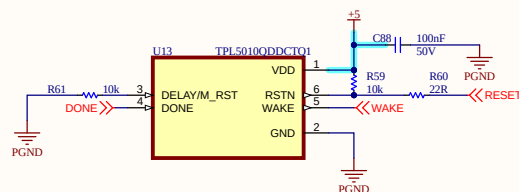
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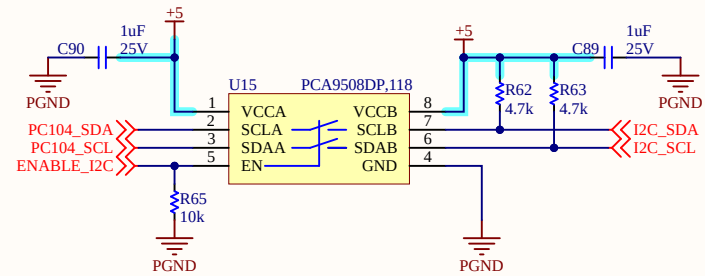
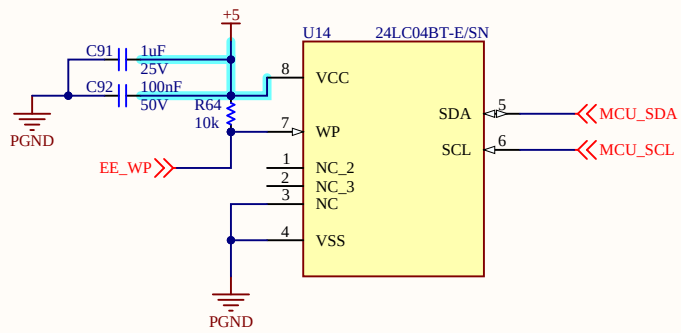
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WATCHDOG

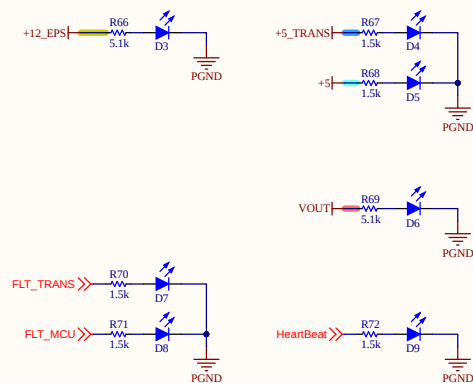


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I2C

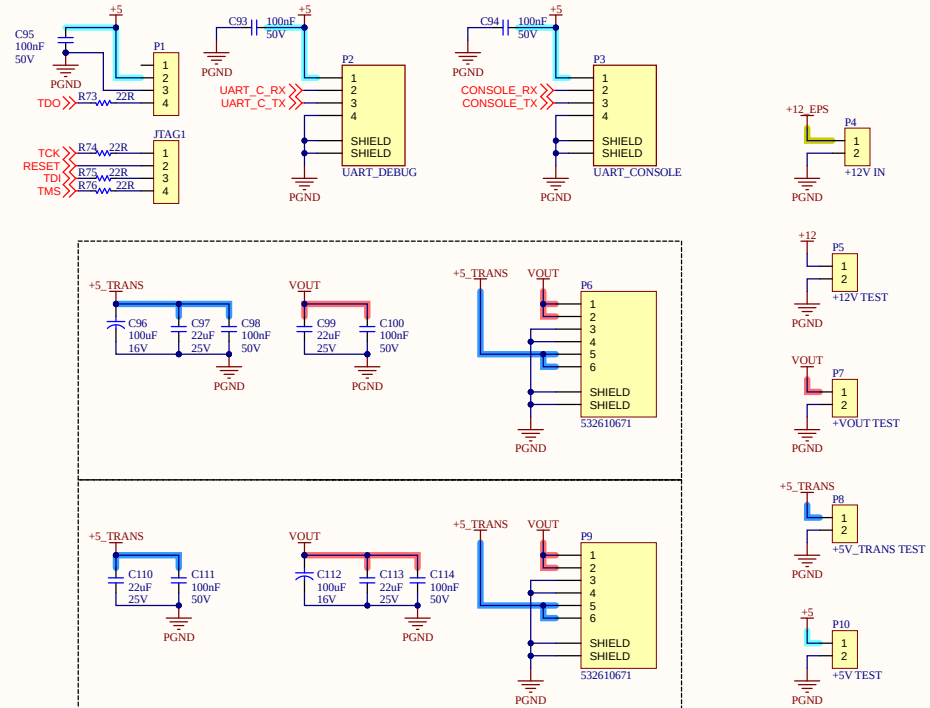
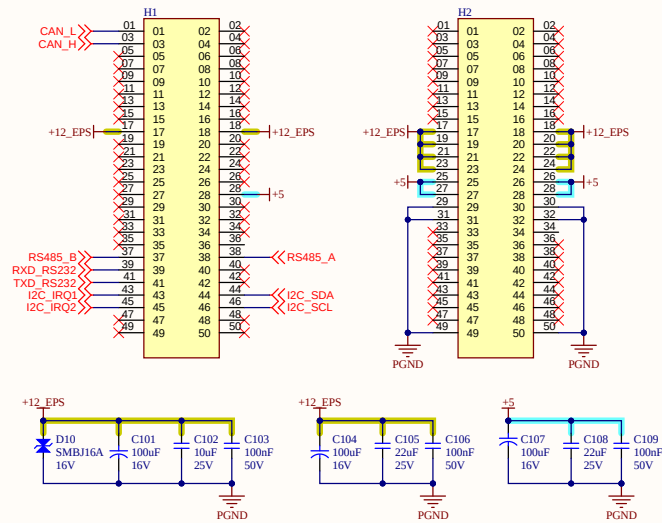


LED



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CONNECTOR



Project: DAL_PDU

Sub-system: IGRVT-50

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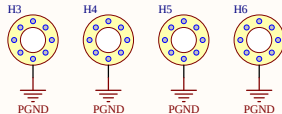
Size: A3

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HOLES



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